Two-body interactions from the tensor-product spectral triple: Selection rules from the algebra, coupling strengths from the metric

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We construct the tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^3}$ for two particles on the Fock-projected three-sphere at finite cutoff $n_{\max}$, and investigate whether the gauged spectral action generates the Coulomb inter-particle interaction from the framework's own algebraic structure. The free tensor-product Dirac $D_{\text{total}} = D_1 \otimes I + \gamma_1 \otimes D_2$ satisfies $\{D, \gamma\} = 0$ exactly, so D^2_{total} factorizes and the free spectral action contains no interaction (Theorem 1). Gauging via the inner fluctuation $A = \sum_i a_i^\dagger [D, a_i]$ from the truncated operator system $\mathcal{O}_{n_{\max}}$ produces a genuine two-body interaction: the connected (non-factorizable) fraction of $\{D, A\}$ is 77% at $n_{\max} = 2$ and 32% at $n_{\max} = 3$ (Proposition 3). The angular decomposition of this interaction is 100% m-conserving, 100% Gaunt-compatible, and pure $k = 0$ monopole at both cutoffs (Theorem 4). However, the radial matrix element ratios do not match the Coulomb $1/r_{12}$ kernel: the coefficient of variation exceeds 1600% and the Pearson correlation is 0.41–0.58 (Proposition 6). These results precisely delineate Paper 31's universal/Coulomb partition at the two-body level: the *algebra* determines which channels couple (angular selection rules); the *metric* determines how strongly (radial coupling strengths). The multi-focal composition wall identified across six independent observables in the GeoVac programme is structurally located at this algebra/metric boundary.

I. INTRODUCTION

The GeoVac framework constructs a discrete spectral triple $(\mathcal{A}, \mathcal{H}, D)$ on the Fock-projected three-sphere [1, 8]: the algebra $\mathcal{A}$ of truncated multipliers on S^3, the Hilbert space $\mathcal{H}$ of Fock-projected states, and the Camporesi–Higuchi Dirac operator D. This single-particle spectral triple has been shown to converge to the continuum S^3 in the van Suijlekom state-space Gromov–Hausdorff distance [10], and the spectral action $\operatorname{Tr} f(D^2/\Lambda^2)$ reproduces the correct gravity sector at one loop [11].

A central open question is whether the framework can derive *inter-particle interactions* from its own structure, rather than importing them as external input. The electron–electron Coulomb repulsion $V_{ee} = 1/r_{12}$, the nucleon–nucleon force, and the core–valence interaction in molecular systems are all two-body couplings that the framework currently imports from standard quantum mechanics. The "multi-focal composition wall" [9] — the observation that the framework couples discrete quantum numbers cleanly but has no native composition theorem for spatial focal lengths — has been documented across six independent observables [16].

In this paper we attack the problem directly. We construct the tensor-product spectral triple $\mathcal{T}_1 \otimes \mathcal{T}_2$ for two particles, each on a copy of the truncated S^3, and compute the gauged spectral action to determine what interaction, if any, emerges from the algebra's inner fluctuations.

The results are sharp: the algebra produces the correct angular selection rules (Gaunt compatibility, m-conservation, monopole dominance), but the radial coupling strengths are not proportional to the Coulomb kernel. This places the multi-focal wall precisely at the boundary between the algebra $\mathcal{A}$ (which determines angular structure) and the Dirac operator D (which determines radial couplings) — the universal/Coulomb partition of Paper 31 [7], now verified at the two-body level.

II. SETUP

A. Single-particle spectral triple

We work with the scalar sector of the Fock-projected S^3 at cutoff $n_{\max}$. The Hilbert space is

$$\mathcal{H}_{n_{\max}} = \operatorname{span}\{|n, l, m\rangle : 1 \le n \le n_{\max}\}, \tag{1}$$

with $\dim \mathcal{H}_{n_{\max}} = N = \sum_{n=1}^{n_{\max}} n^2$. For the Dirac structure, we chirality-double to $\hat{\mathcal{H}} = \mathcal{H}^+ \oplus \mathcal{H}^-$ with $\dim \hat{\mathcal{H}} = 2N$.

The Dirac operator is the Camporesi–Higuchi form [12]:

$$D|n, l, m, \chi\rangle = \chi \left(n + \tfrac{1}{2}\right) |n, l, m, \chi\rangle, \tag{2}$$

where $\chi = \pm 1$ is the chirality. The chirality operator γ swaps the two sectors: $\gamma|n, l, m, +\rangle = |n, l, m, -\rangle$.

The algebra $\mathcal{A} = \mathcal{O}_{n_{\max}}$ is the Connes–van Suijlekom [14] truncated operator system, generated by the 3-Y multiplier matrices M_{NLM} with entries

$$(M_{NLM})_{(nlm),(n'l'm')} = \int_{S^3} \overline{Y^{(3)}_{nlm}} \cdot Y^{(3)}_{NLM} \cdot Y^{(3)}_{n'l'm'} \, d\Omega. \tag{3}$$

These are embedded into $\hat{\mathcal{H}}$ as $\hat{M} = \operatorname{diag}(M, M)$, acting identically on both chirality sectors.

B. Tensor-product Dirac

For two particles, the total Hilbert space is $\hat{\mathcal{H}}_1 \otimes \hat{\mathcal{H}}_2$ with dimension $(2N)^2$. The tensor-product Dirac operator is the standard product [13]:

$$D_{\text{total}} = D_1 \otimes I_2 + \gamma_1 \otimes D_2. \tag{4}$$

The chirality twist γ_1 ensures D_{total} is a first-order operator on the graded tensor product.

III. FREE FACTORIZATION

Theorem 1 (Free factorization)**.** *For the Camporesi–Higuchi Dirac* (2) *on the chirality-doubled scalar basis,* $\{D, \gamma\} = 0$ *exactly. As a consequence,*

$$D^2_{\text{total}} = D_1^2 \otimes I_2 + I_1 \otimes D_2^2, \tag{5}$$

and the spectral action $\operatorname{Tr} f(D^2_{\text{total}}/\Lambda^2) = \operatorname{Tr} f(D_1^2/\Lambda^2) \cdot \operatorname{Tr} f(D_2^2/\Lambda^2)$ *factorizes exactly at every cutoff* Λ.

Proof. The Dirac eigenvalue on $|n,l,m,\chi\rangle$ is $\chi(n+\frac{1}{2})$. The chirality operator sends $\chi \to -\chi$. For any basis state,

$$(D\gamma + \gamma D)|n,l,m,\chi\rangle = \chi(n+\tfrac{1}{2})(-\chi)(n+\tfrac{1}{2}) + (-\chi)(n+\tfrac{1}{2})\chi(n+\tfrac{1}{2}) = 0,$$

using $D|n,l,m,-\chi\rangle = -\chi(n+\frac{1}{2})|n,l,m,-\chi\rangle$. Hence $\{D,\gamma\} = 0$. Expanding D^2_{total}:

$$D^2_{\text{total}} = D_1^2 \otimes I + (D_1\gamma_1 + \gamma_1 D_1) \otimes D_2 + I \otimes D_2^2.$$

The cross term vanishes by $\{D_1, \gamma_1\} = 0$. □

Remark 2. *This is the correct physical statement: free particles do not interact. The spectral action of the free Dirac gives independent single-particle kinetic energies. Any interaction must enter through* gauging.

Numerical verification. At $n_{\max} \in \{2,3\}$, the heat trace ratio $\operatorname{Tr}(e^{-tD^2_{\text{total}}})/[\operatorname{Tr}(e^{-tD_1^2}) \cdot \operatorname{Tr}(e^{-tD_2^2})] = 1.000000$ to machine precision at all tested $t \in \{0.1, 0.5, 1.0, 2.0\}$.

IV. GAUGED INTERACTION

The inner fluctuation of D_{total} by the tensor-product algebra $\mathcal{A}_1 \otimes \mathcal{A}_2$ is

$$[D_{\text{total}}, \hat{M}_i \otimes \hat{M}_j] = [D_1, \hat{M}_i] \otimes \hat{M}_j + \gamma_1 \hat{M}_i \otimes [D_2, \hat{M}_j]. \tag{6}$$

Both terms are generically nonzero: $[D, \hat{M}]$ is nonzero whenever $\hat{M}$ has cross-shell matrix elements (coupling states with different n, hence different D-eigenvalues). At $n_{\max} = 2$, four of 14 multiplier generators have nonzero commutators; at $n_{\max} = 3$, 29 of 55.

TABLE I. Heat trace ratio $\operatorname{Tr} e^{-t(D+\varepsilon A)^2}/[\operatorname{Tr} e^{-tD_1^2} \cdot \operatorname{Tr} e^{-tD_2^2}]$ at $t = 0.5$ for the full-algebra gauge field.

ε	$n_{\max} = 2$	$n_{\max} = 3$
0.0	1.000000	1.000000
0.1	0.995465	0.838438
0.5	0.977815	0.410165
1.0	0.956832	0.185405

The *rotationally invariant* one-form is constructed using the conjugate pairing [13]:

$$A = \sum_{(NLM)} \sum_{(N'L'M')} \left(\hat{M}^\dagger_{NLM} \otimes \hat{M}^\dagger_{N'L'M'}\right) \cdot [D_{\text{total}}, \hat{M}_{NLM} \otimes \hat{M}_{N'L'M'}], \tag{7}$$

summing *independently* over all allowed multiplier labels on each tensor factor in $\mathcal{O}_{n_{\max}}$. This is the full inner fluctuation of the product algebra $\mathcal{A} \otimes \mathcal{A}$, not only the diagonal ($N'L'M' = NLM$) terms; all numerical results in this paper use this double sum. The dagger pairing ensures m-conservation (total magnetic quantum number $M_1 + M_2$ is preserved) and Hermiticity of A.

Proposition 3 (Connected interaction)**.** *The anti-commutator* $\{D_{\text{total}}, A\}$ *contains a non-factorizable (connected) two-body component. Define the connected part via partial-trace subtraction:*

$$V_{\text{conn}} = \{D,A\} - \frac{1}{d_2}\operatorname{Tr}_2\{D,A\} \otimes I_2 - I_1 \otimes \frac{1}{d_1}\operatorname{Tr}_1\{D,A\} + \frac{\operatorname{Tr}\{D,A\}}{d_1 d_2} I_1 \otimes$$

At $n_{\max} = 2$*:* $\|V_{\text{conn}}\|/\|\{D,A\}\| = 76.7\%$*. At* $n_{\max} = 3$*:* $\|V_{\text{conn}}\|/\|\{D,A\}\| = 32.4\%$.

The interaction is genuine: the connected fraction is nonzero, and the heat trace $\operatorname{Tr} e^{-t(D+\varepsilon A)^2}$ departs from the factorized value monotonically with ε (Table I).

V. ANGULAR SELECTION RULES

Theorem 4 (Angular structure of the connected interaction)**.** *The connected two-body interaction* V_{conn} *from the rotationally invariant gauge field* (7) *satisfies:*

(i) m***-conservation:*** $m_1 + m_2 = m_1' + m_2'$ *for all nonzero matrix elements — structural for all* n *(the* $M^\dagger \cdot [D,M]$ *conjugate pairing forces the gauge field's total* $M = 0$*), confirmed numerically at* $n_{\max} \in \{2,3\}$.

(ii) ***Gaunt compatibility:*** $|\Delta l_1| = |\Delta l_2| = k$ *for all nonzero* $(l_1,l_2) \to (l_1', l_2')$ *channels — structural for all* n *via the* 3*-*Y *selection rules of the multiplier algebra (consistent with the potential-independent angular sparsity theorem of Paper 22 [6]), confirmed at* $n_{\max} \in \{2,3\}$.

(iii) ***Pure monopole:*** 100% *of* $\|V\|^2$ *is in the* $k = 0$ *multipole channel,* verified at $n_{\max} \in \{2, 3\}$ *(computational; not asserted beyond the tested cutoffs).*

(iv) ***Hermiticity and exchange symmetry*** *hold to machine precision* ($< 10^{-17}$).

Verification. Direct computation from the 4-index tensor $V_{\text{conn}}[a, b, c, d]$ extracted from the positive-chirality sector at $n_{\max} = 2$ ($N = 5$, 25 nonzero elements) and $n_{\max} = 3$ ($N = 14$, 196 nonzero elements). The m-conservation is enforced by the $M^\dagger \cdot [D, M]$ conjugate pairing, which ensures the total M quantum number of the gauge field is zero. The Gaunt compatibility follows from the 3-Y selection rules of the multiplier algebra. All four items (m-conservation, Gaunt compatibility, 100% $k = 0$ monopole, and the connected-fraction proposition above) are regression-pinned in `tests/test_paper31_two_body.py` (`test_spectral_action_two_body_nmax2` and `..._nmax3_decreasing`), which also verifies the radial-vs-Coulomb Pearson decreasing with $n_{\max}$. □

Remark 5. *The* $k = 0$ *monopole channel corresponds to the* F^0 *Slater integral in the Neumann expansion of* $1/r_{12}$ *(Paper 12 [2]). Higher multipoles* $k = 1, 2, \ldots$ *would appear at larger* $n_{\max}$ *or in the full spinor sector. The* absence *of the* $k = 1$ *dipole is correct for identical particles by parity.*

VI. RADIAL COMPARISON WITH COULOMB

Proposition 6 (Radial weights do not match Coulomb)**.** *The matrix element ratios* $V_{\text{conn}}[a, b, c, d]/\langle ab|r_{12}^{-1}|cd\rangle$ *are not constant across the nonzero elements. At* $n_{\max} = 2$*: coefficient of variation* CV = 16.5*, Pearson correlation* $r = 0.58$*, sign agreement* 68%*. At* $n_{\max} = 3$*:* CV = 20.3*,* $r = 0.41$*, sign agreement* 54%*.*

The spectral-action interaction is a strict *subset* of the Coulomb matrix elements: all nonzero spectral elements also appear in the Coulomb matrix (0 spectral-only entries), but the Coulomb matrix has many additional nonzero elements (240 at $n_{\max} = 2$, 15 168 at $n_{\max} = 3$) that are zero in the spectral interaction. The Pearson correlation *decreases* with $n_{\max}$, indicating this is structural rather than a finite-size artifact.

Remark 7. *The* $1/r_{12}$ *Coulomb kernel in the Fock-projected basis arises from the chordal distance identity under stereographic projection (Paper 7 [1]). This is a* metric *property of* S^3*, not a spectral-action property. The spectral action* $\operatorname{Tr} f(D^2/\Lambda^2)$ *computes heat-kernel coefficients (Seeley–DeWitt), while* $1/r_{12}$ *is a Green's function kernel — fundamentally different objects at the mathematical level.*

VII. THE ALGEBRA/METRIC BOUNDARY

The results of Sections V–VI precisely delineate the universal/Coulomb partition of Paper 31 [7] at the two-body level:

- The **algebra** $\mathcal{A}$ (truncated operator system, 3-Y multipliers) determines the angular selection rules: Gaunt compatibility, m-conservation, and the multipole hierarchy. These are *universal* — they hold for any central potential, as proved by the angular sparsity theorem of Paper 22 [6].

- The **Dirac operator** D (Camporesi–Higuchi eigenvalues) determines the radial coupling strengths. The gauged spectral action generates an interaction whose radial weights depend on the specific spectrum of D, not on the $1/r$ Coulomb potential. The Coulomb kernel is additional input from the Fock projection's metric structure.

This decomposition explains the "multi-focal composition wall" documented across six observables in the GeoVac programme [9]: the wall sits at the $\mathcal{A}/D$ boundary. The framework couples discrete labels (angular quantum numbers) via the algebra's inner fluctuations, but the spatial coupling strength (radial integrals) requires the metric — the Fock projection that maps the graph onto S^3, and with it, the $1/r$ potential.

Corollary 8. *For two particles on* $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^3}$*, the interparticle interaction splits as*

$$V_{12} = \underbrace{\textit{angular selection rules}}_{\textit{from } \mathcal{A}} \times \underbrace{\textit{radial coupling strengths}}_{\textit{from } D/\textit{metric}}. \tag{8}$$

The first factor is framework-native (derived from inner fluctuations). The second factor is calibration data (the Slater integrals R^k *of Paper 12 [2]).*

VIII. IMPLICATIONS

A. For atomic and molecular structure

The result vindicates the composed-geometry architecture of Paper 17 [4]: the Gaunt selection rules that give the exactly Q-linear composed Pauli count ($N_{\text{Pauli}} = 27.90 \times Q$ at fixed basis) of Paper 14 [3] are not ad hoc — they emerge from the tensor-product algebra's inner fluctuations. The radial Slater integrals that complete the Hamiltonian are the irreducible calibration content, consistent with Paper 18's exchange-constant taxonomy [5] (calibration tier).

B. Connection to double factorization and tensor hypercontraction

The two-electron operator decomposition $V_{ee} = \sum_P L_P \otimes L_P$ used in double factorization [17] and tensor hypercontraction [18] is literally an element of $\mathcal{A} \otimes \mathcal{A}$ in the tensor-product spectral triple constructed here. In the GeoVac basis, this decomposition is supplied analytically by the multipole expansion

$$\frac{1}{r_{12}} = \sum_{L,M} \frac{4\pi}{2L+1} \frac{r_<^L}{r_>^{L+1}} Y_{LM}(\Omega_1)\, Y_{LM}^*(\Omega_2),$$

with L_P proportional to Y_{LM} and rank equal to the number of distinct radial-density channels $\rho_\alpha(r) = R_{n_a l_a}(r)\, R_{n_b l_b}(r)$ at each angular momentum L.

Numerical double factorization applied to a GeoVac composed ERI tensor reproduces this analytic decomposition bit-exactly at the production basis: the relative reconstruction error $\|G - VKV^\top\|_F/\|G\|_F$ between the empirical ERI tensor and its multipole-VKV form is 10^{-16} at $n_{\max} = 2$ and 10^{-14} at $n_{\max} = 3$, i.e. machine precision. Every singular vector of the reshaped two-electron tensor lives strictly inside the multipole-channel subspace at every basis size tested ($n_{\max} \in \{2, 3, 4\}$), with total subspace overlap unity to machine precision. At the production basis ($n_{\max} = 2$, valence $\{1s, 2s, 2p\}$), the double-factorization rank equals the analytic distinct-radial-density count exactly — seven, distributed as four densities at $L = 0$, two at $L = 1$, one at $L = 2$.

At larger basis ($n_{\max} \geq 3$), the radial-integral matrix $R^L(\rho_\alpha; \rho_\beta) = \langle \rho_\alpha | r_<^L / r_>^{L+1} | \rho_\beta \rangle$ develops a graded singular-value spectrum that decays over many orders of magnitude rather than terminating at a clean cutoff. This is the well-known Laguerre-basis ill-conditioning of hydrogenic ERIs — the same phenomenon that motivates resolution-of-identity and Cholesky methods in standard atomic and molecular structure. Radial densities with similar effective exponents $(1/n_a + 1/n_b)/2$ produce nearly proportional contributions against the Coulomb kernel, so the apparent "rank" of R^L depends on threshold. The additional compression that DF finds beyond the bare analytic multipole count is therefore not a new algebraic structure but a numerical realization of the same spectral content, exposed through SVD ordering. Operationally: double factorization on a GeoVac Hamiltonian is the multipole expansion accessed by SVD rather than by closed form. This is a numerical observation about how DF orders an already-established spectral content (the analytic multipole count, whose $k = 0$ monopole dominance is pinned in `tests/test_paper31_two_body.py`), not a separate algebraic claim.

C. For nuclear structure

The same $\mathcal{A}/D$ split applies to the nucleon–nucleon interaction. The Minnesota potential's angular structure (central force, spin-dependent channels) is algebra-side; the radial Gaussian form factors ($V_R e^{-\kappa_R r^2}$, etc.) are D-side calibration. This explains why the nuclear model requires imported potentials: the framework derives *which* nuclear channels couple but not the NN force strength.

D. For gravity

The gravity sector (Paper 51 [11]) is special: the spectral action directly gives metric properties (Einstein–Hilbert action, Newton's constant, cosmological constant) because gravity *is* the metric. The two-body Coulomb interaction is a Green's function kernel, not a heat-kernel coefficient, so it lives on the other side of the $\mathcal{A}/D$ boundary. This structural difference explains why gravity results are bit-exact while inter-particle interactions require imported potentials.

IX. DISCUSSION

The tensor-product spectral action generates genuine two-body physics: a non-factorizable interaction with the correct angular selection rules. That the radial weights do not match Coulomb is not a failure of the framework but a structural theorem about *what spectral actions can compute*: heat-kernel coefficients (metric curvature, not Green's functions). The Coulomb potential is the Green's function of the Laplacian, $\nabla^2 G = -4\pi\delta^{(3)}$; it requires solving a differential equation on the manifold, not tracing over the spectrum.

The practical consequence is that inter-particle radial integrals (Slater R^k, Minnesota form factors, Clementi–Raimondi screening) are irreducible calibration data in the GeoVac framework. The angular selection rules that govern which integrals are nonzero are framework-native. This is a more precise statement of the multi-focal composition wall: it is not a missing tool but the structural boundary between what the algebra determines and what the metric determines in a spectral triple.

Open question. Whether the Green's function kernel can be recovered from the spectral triple by a construction other than the spectral action — for example, via the resolvent $(D - z)^{-1}$ or the Connes distance function — remains an open problem. The resolvent naturally gives the propagator $\langle x | (D-z)^{-1} | y \rangle$, which in the continuum limit reproduces the Green's function. A resolvent-based derivation of $1/r_{12}$ from the tensor-product spectral triple would close the multi-focal wall.

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